

Mobile heritage prototype

Canal Quest at Chang Gate

A playful web app that turns a passive heritage walk into a clear loop of exploring, noticing, completing tasks, collecting stamps, and creating a memory.

CPT208 Human-Centric Computing

Group: [A1-1]

Xinrong Li 2363123 Junjie Wang 2360859
Huangxin Li 2364485 Sihang Sha 2363107

No app install Mobile first Playful tasks

Project Aim

Design opportunity

Chang Gate is rich in history, but onsite visits can become short, passive, and photo-only. Canal Quest reframes the visit as a lightweight self-guided journey: visitors open the website, choose nearby stops, respond to small observation prompts, and receive visible progress after each action.

Active noticing Self-paced route Collectable progress Memory making Browser-based access

18–30

Primary young visitor
and student audience

6

Prototype states in the
main app flow

0

Installation steps before
starting

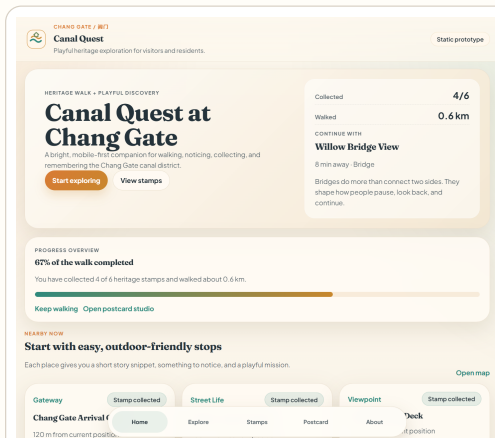
Mobile application



Online poster preview

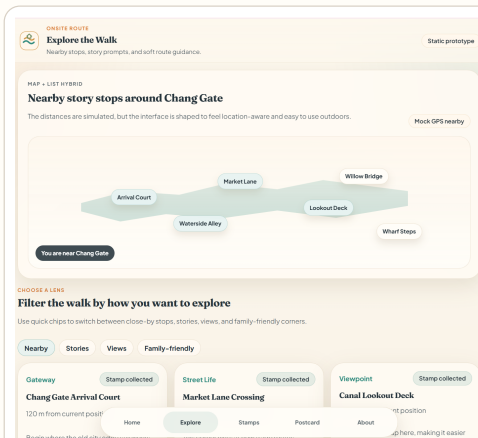


Application Flow: From Arrival to Memory



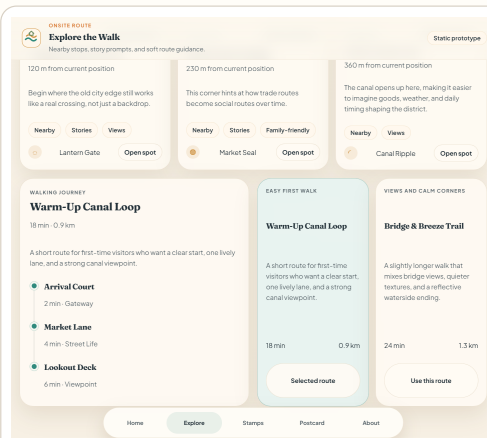
01
Start the walk

Clear entry, progress status, and next stop



02
Choose a stop

Map-like overview and route choices



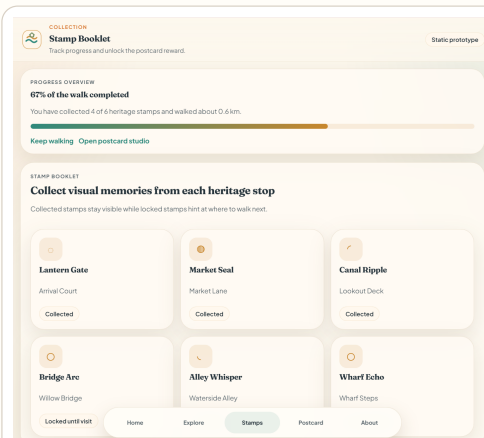
03
Complete a task

Short prompts linked to site details



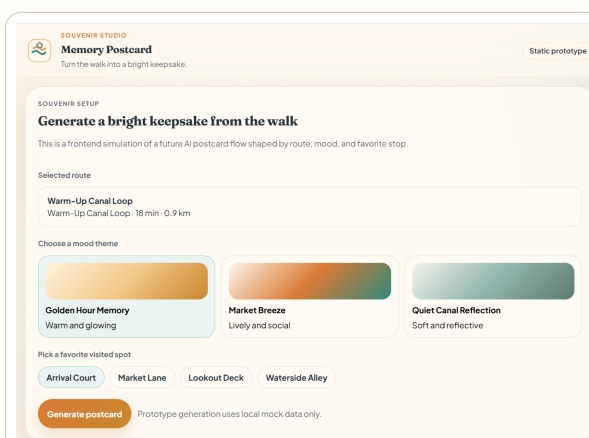
04
Notice the place

Gallery records real visual moments



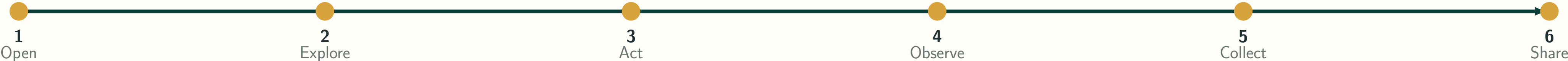
05
Collect reward

Stamp booklet makes progress visible



06
Create memory

Postcard closes the visit with reflection



The poster foregrounds the product flow. The gallery screen now carries the place-based visual evidence inside the app itself.

User Problem and Design Goals

What current visits miss

- Heritage visits often stop at reading boards, taking photos, and leaving.
- Long explanations compete with short attention spans during walking.
- Visitors need guidance without feeling forced into a strict tour.
- Younger users respond better to visual, mobile, and progress-based interaction.

Human-centered goals

- Keep the first action obvious on a phone screen.
- Use short tasks to make visitors look at real details around them.
- Reward progress without creating pressure or complicated rules.
- Let the visit end with a personal output rather than only information.

Positioning

A lightweight web app, not a heavy AR system

The prototype favors accessibility, speed, and feasibility for a short onsite walk.

Tone

Playful but calm

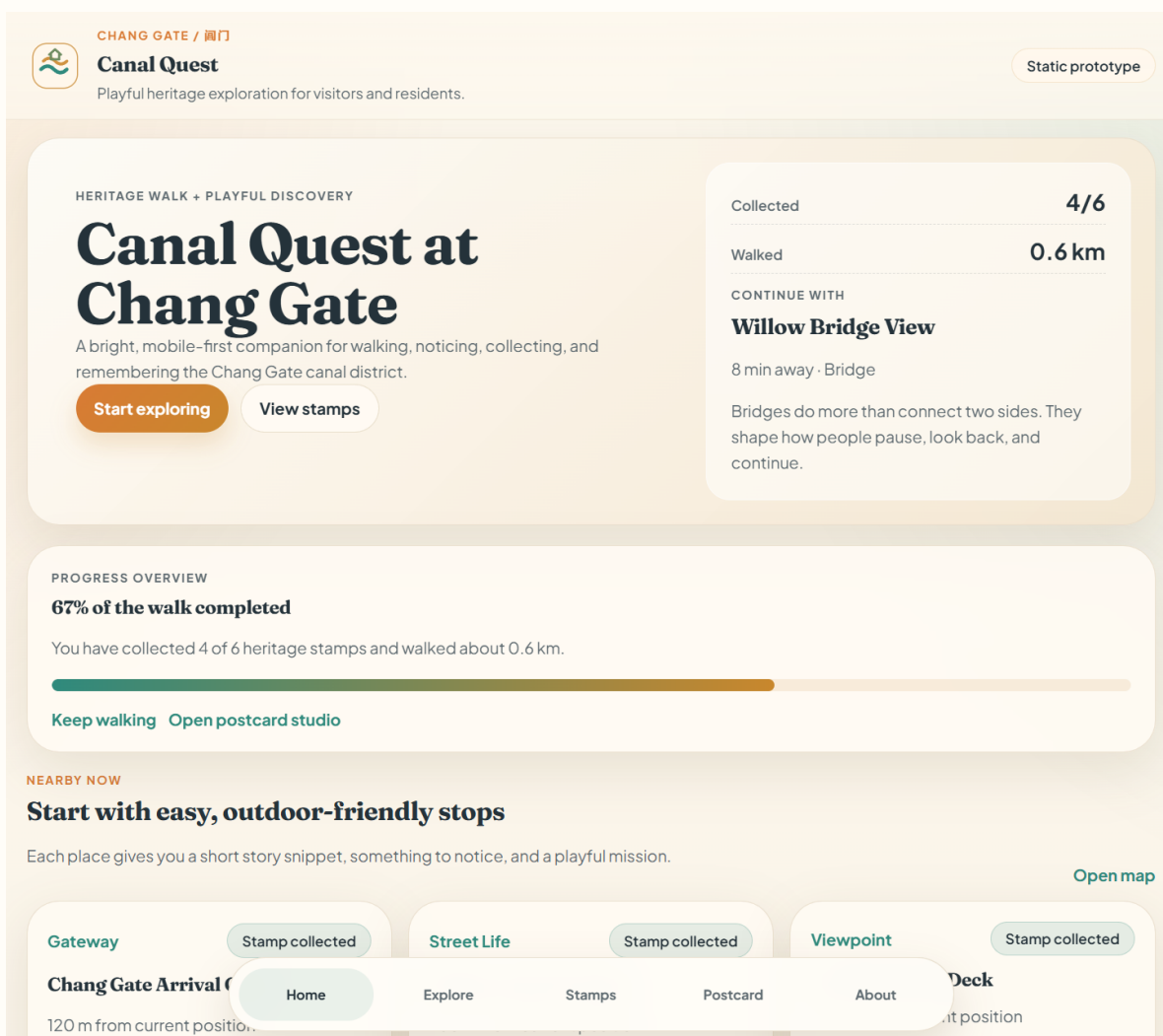
The interface uses soft colors, clear cards, and small rewards to support exploration without distracting from the site.

Design principle

Low pressure, high guidance

The app should make visitors confident about what to do next while still leaving room for self-paced walking and casual discovery.

Main Interface Evidence



Homepage role

The home page acts as the control center: it introduces the walk, shows progress, lists the next stop, and keeps the first actions visible.

Visible status

Stamp count, distance, and completion state make the route feel continuous.

Low barrier

Clear buttons reduce setup time and make entry quick.

Place aware

Stops are connected to real locations around Chang Gate.

Return paths

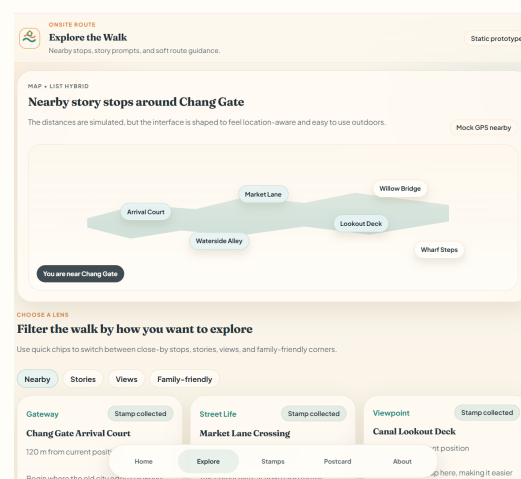
Bottom navigation keeps the main screens within reach.

Screen hierarchy

Start action first, supporting detail second

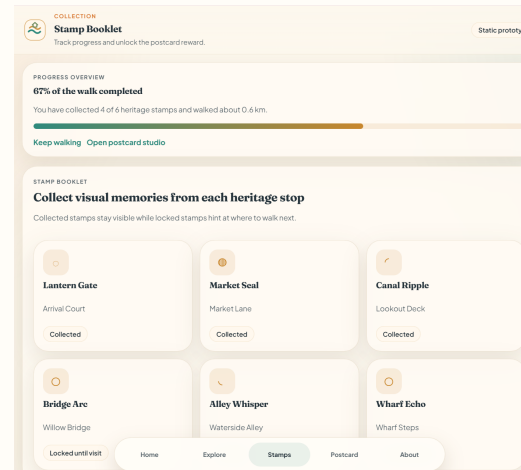
The homepage keeps the route status, next location, and primary buttons in the same visual field so the user can begin without reading a long guide.

Interaction System



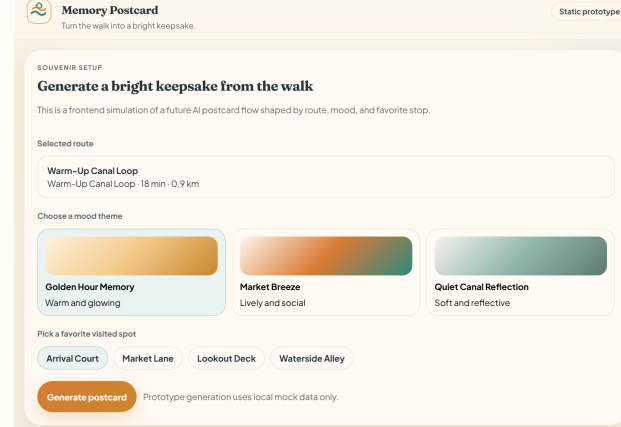
Explore

Nearby stops, filters, and route guidance



Stamps

Collected rewards and completion feedback



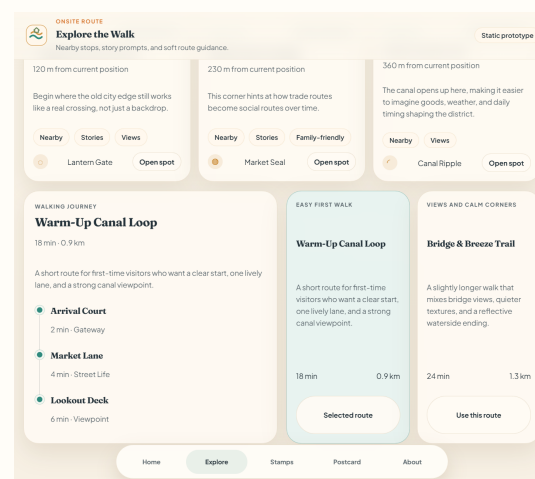
Postcard

A shareable memory output

Core mechanism

Each screen supports one part of the same loop: choose where to go, do a small action, receive feedback, and continue. The interaction stays simple enough for outdoor use while still giving the visit a sense of progress.

Explore Prompt Reward Reflect



Task

Observation prompts tied to the place

Gallery and Heritage Connection



Why the scenic photos were removed

The gallery page already presents Chang Gate imagery as part of the product. Using it avoids a separate scenic-photo section and keeps the poster focused on the application.

Design value

- Visual proof of the real heritage site
- Stronger link between app content and walking context
- Less duplicated material on the poster
- More space for the interaction flow

Poster decision

The scenic photos are still available in the folder, but the final poster uses app screens only so every visual element explains the prototype.

Evaluation, Impact, and Next Steps

Early findings

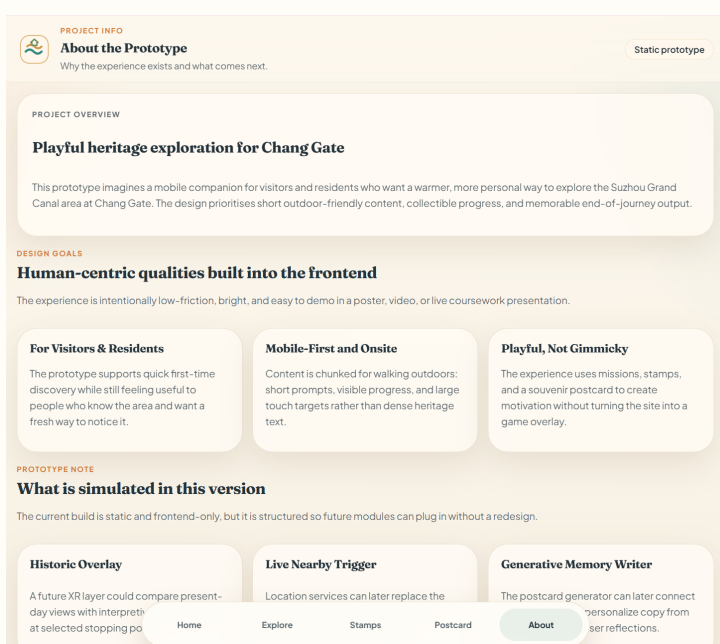
- The concept is easy to understand from the first screen.
- Stamp collection makes the walk feel more motivating.
- Short prompts are more suitable than long text blocks outdoors.
- A browser prototype lowers the entry barrier for casual visitors.

Planned improvements

- Sharper map and route guidance for real onsite use.
- More varied mission types beyond observation prompts.
- Stronger Chang Gate identity in icons, stamps, and copy.
- More user testing with visitors unfamiliar with the site.

Human-centric relevance

Canal Quest supports inclusive access through a familiar mobile browser, makes local culture participatory, and gives visitors a lightweight reason to slow down, notice details, and remember the walk.



About / Prototype rationale

This screen explains the concept, simulated features, and human-centric qualities.

References and final group details should be inserted before submission according to the module requirements.